Recirculation: Spectral Ranking of Journals and Organisations in Economic and Business Science

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Abstract

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1 Introduction

1.1 Social Systems Theory of Citation

1.2 Citation Matrices

Pinski and Narin (1976) define the citation matrix C_{ij} as a table of *references* from unit i to unit j or equally of *citations* to unit j from unit i. We follow this convention with i, j denoting rows, columns respectively, and note that the transposed convention is also frequently adopted (e.g. Prathap, 2019).

In this paper we consider individual works aggregated into journals. Each paper has an affiliation list comprising authors and their institutions, as well as a list of references to earlier works. The references establish directional reference/citation links between works, and hence between journals, authors and institutions. When aggregated, these links become weights in the citation matrix.

There are many ways to assemble the aggregations. We adopt the viewpoint that a single reference has unit value to be divided among referencing paths. A weight of 0.5 is allocated to the path from referencing journal to cited journal, while the other 0.5 is allocated equally across the paths between institutions. If the referencing work has N^r distinct referencing institutions and the cited work N^c cited institutions, each institution/institution path receives a weight of $0.5/(N_i^r \times N_j^c)$. These rules lead to a citation matrix that has two blocks, one for journals and one for institutions. Mixed journal/institution paths could be included in the citation matrix; instead we recirculate between the blocks as described below.

1.3 Measures of Influence

1.4 Eigenvectors and Power Iterations

Pinski and Narin (1976)

The long-standing, influential, and often criticised ISI journal impact factor introduced in Garfield (1955) reports the average number of citations per article in a journal. The adjustable features of the JIF include the census and target time intervals, the journal set, and the rule for including works from these journals as citation sources. The JIF says nothing about the status of the citing works.

Many researchers have advocated alternatives to the JIF that take account of properties of the citing sources. The Audience Factor (Zitt and Small, 2008) adjusts for "disciplinary" differences (in practice, "journal-level" differences) by boosting or diminishing the weight of a citation in proportion the fraction of all citer-side references that appear in the journal under consideration. This accounts for average disciplinary differences in the length of reference lists and in the delay before work is cited, both factors that influence the JIF. Like the JIF, the AF says nothing about the status of the citing works.

Early work by Pinski & Narin (Pinski and Narin, 1976) presented an approach to journal influence that accounts for the influence a journal when it provides citations to the journal of interest. The approach leads to an iteration over all journals to find the influence of each journal. Geller reported a subtle mathematical correspondence between Pinski-Narin and a Markov chain; this line of work has continued, but is not addressed here (Vigna, 2016). Resonating with the Pinksi-Narin approach, social network theorists computed graph-based measures of social status that start from a perspective like that of PN, where status depends not only on how many people know you, but also on the status of those people. This work includes Bonacich's Network Eigenvector Centrality (Bonacich, 1987) and Kleinberg's Network Hubs and Authorities (Kleinberg, 1999).

The Pinski & Narin work was acknowledged in the original PageRank patent (Page, 1998) and cleverly adapted in 1998 by Brim and Page (Brin and Page, 1998) to become the Google PageRank algorithm used to explore the influence of web-page links. The most obvious counterpart to webpage links is article-to-article reference/citation links, with each reference or citation having unit weight. We are interested here in "units" that are assemblies of articles, in particular journal-to-journal and journal-to-institution links. The assembly a journal or institution corpus from its articles implies that unit links should add all the relevant individual article-to-article citations, so that unit links become weighted, unlike URLs. Unlike web links, citation links also have a time direction, where a later reference is a citation to an earlier paper, so that the relationship is not symmetric (Wouters, 1999). Thus, the standard PageRank algorithm is not directly useful, but several variants have been developed.

Bollen et al (2006) advocated an extension of PageRank as an alternative to the JIF to include the "prestige" of citing journals into the calculation of a journal's influence (they called this "status"). Modifying the single link of web pages, Bollen et al. used the count of all citations as in Pinski & Narin, and dubbed the revised algorithm weighted PageRank. Parenthetically, they also introduced the composite JIF/PageRank indicator as the geometric mean of the two indicators.

The work of Pinski & Narin and Bollen et al. was developed further to construct the Eigenfactor Score by Bergstrom et al (2008) and the related Article Influence Score Franceschet (2011).

The Reputable Citation algorithm (RC) by Safon, Docampo and Cram (Safón et al, 2025) is an extension of this class of algorithms. Recognising that influence indicators based on citations alone are subject to compromise through manipulated reference lists, RC applies a prior influence or prestige weight to each journal, derived from exogenous indicators of the prestige of the institutions that publish in it.

The remainder of the article is organized as follows. In Section 2, we cover the eigenvector-based methods' mathematical background to introduce the mathematical formulation of the JIRC algorithm in Section 3. Section 4 covers the data and methods used to compute JIRC scores. In Section 5, we present the results of our analysis of the ESI field of Economics & Business for journals and institutions, and their impact on the individual screening of researchers. Section 6 is devoted to a discussion of the results, while Section 7 closes the paper by presenting our conclusions.

2 Eigenvector based methods

We examine in this section the mathematical background of the Pinski & Narin (PN) principal eigenvector method to later introduce the modifications leading to the new JIRC algorithm.

2.1 The citation matrix

Following PN, let's define the "unit" as a journal j which is a member of a set of selected journals $\mathbf{J}$, where $j = 1 \dots M$.

Let $C = (c_{ij})$ be a matrix whose elements are the sum of references emitted by the articles published in journal i (rows) that become citations received by any of the articles published in journal j (columns) during the target publication window. If we ignore journal self-citations, a common practice, then $c_{ii} = 0$. Define the sum of all references from i as

$$s_i = \sum_{j=1}^{M} c_{ij} \tag{1}$$

The matrix C can be normalised as the row stochastic matrix P where

$$P = (p_{ij}) = (c_{ij}/s_i) \tag{2}$$

in which case citations from a journal are normalized by the total number of citations coming from that journal to the other journals in the ensemble.

Domingo, this text is correct but I did not previously notice that it is not the PN normalisation. The PN matrix P is unlikely to be stochastic, explaining why Geller (her equation 3) introduces a kind of "dual" to the PN matrix to arrive at a Markov interpretation. Vigna p 442 refers to the PN normalisation as "slightly bizarre". This helps explain the PN influence-per-article step, as well as the move from the Eigenvector score of a journal to the article influence score or the similar step taken by Bollen etal. The PN normalisation is to the number of references coming from journal i:

$$P = (p_{ij}) = (c_{ij}/s_j) \tag{3}$$

This is total number of citations to i - or the total number of references from other units - divided by total number of references from i – i.e. input references/ output references. This is not the same as the total number of citations to i divided by the total number of citations from i, which is the recipe for a stochastic matrix

A stochastic matrix is a transition probability Markov matrix. The repeated product P^n represents a step-wise chain from an initial to a final state after n steps. If the matrix P is irreducible, there is a single asymptotic final state; if not, there is no unique final state. It is desirable for computational reasons that the stochastic matrix P is irreducible. We will assume that in what follows.

2.2 Influence Factor (Pinski & Narin - PN)

Define the journal influence weight vector $w = (w_j)$ as a column vector of influence weights for each journal in the set. The influence weight of a journal averages the weighted number of citations from every journal to that journal so that, ultimately, journals with "above average" influence weight are those that receive more citations from journals with high influence weight and vice versa. This leads to an implicit equation for w

$$w_j = \sum_{i=1}^{N} p_{ij} w_i \tag{4}$$

$$w = P^T w \tag{5}$$

This equation is not the PN equation. Their equation 1 can be written like this one below - I've used the index j on the left, they use i

$$S_j W_j = \sum_{i=1}^{N} C_{ij} \tag{6}$$

This equation defines the influence weight for a journal. The influence weight per publication w' is

$$w'_j = \frac{(P^T w)_j}{a_j} \tag{7}$$

where a_j is the number of articles published in journal j in the time period selected.
PN uses the iteration (called the "Power method")

$$w(k) = \frac{P^T w^{(k-1)}}{|P^T w^{(k-1)}|} \tag{8}$$

that converges to the principal eigenvalue and the desired influence vector.

3 The new JIRC algorithm

The dataset now contains the ensemble of journals, $j = 1 \ldots M$, as well as a collection of institutions, $i = 1 \ldots N$ with a sizable number of papers in the field and period under analysis.

When computing journal influence scores, JIRC will inject values of citing article prestige, v_i, based on institutions assessments derived from the dataset. Conversely, to computing institutional influence scores, JIRC will inject values of citing article prestige, w_j, based on journal assessments derived from the same dataset. We will formulate the problem as a (re)circulation of reputation. The idea is to go from institutions to institutions through journals and vice versa.
The problem can be stated as follows.

Let's introduce citation matrix A, from institutions ($i = 1 \ldots N$) to journals ($j = 1 \ldots M$), $A = (c_{ij})$, whose elements are the sum of references emitted by the articles published by institution i (rows) that become citations received by any of the articles published in journal j (columns) during the target publication window.
Let's again define the overall sum of all references from institution i as

$$s_i = \sum_{j=1}^{M} c_{ij} \tag{9}$$

The matrix A can be normalised as the row stochastic matrix P_A, where

$$P_A = (a_{ij}) = (c_{ij}/s_i) \tag{10}$$

in which case references from an institution are normalized by the total number of references from that institution to all journals in the dataset.

We also introduce citation matrix B, from journals to institutions, $B = (c_{ji})$, whose elements are the sum of references emitted by the articles published by journal j (rows) that become citations received by any of the articles published in institution i (columns) during the target publication window.

$$s_j = \sum_{i=1}^{N} c_{ji} \tag{11}$$

The matrix B can be normalised as the row stochastic matrix P_B, where

$$P_B = (b_{ji}) = (c_{ji}/s_j) \tag{12}$$

in which case, references from a journal are normalized by the total number of references from that journal to all the institutions in the dataset.

The influence vectors are connected by the two matrices. P_A^T is $M \times N$, whereas P_B^T is $N \times M$. Both influence column vectors, w, $M \times 1$, for journals and v, $N \times 1$, for institutions, receive reputation shots from each other, $w = P_A^T v$, $v = P_B^T w$.

Hence, we can now treat the reputation transfer full circle, as

$$v = P_B^T P_A^T v \tag{13}$$
$$w = P_A^T P_B^T w \tag{14}$$

Matrices $P_A^T P_B^T$ and $P_B^T P_A^T$ are both square, although they have, in general, different sizes: $M \times M$ and $N \times N$, respectively.

However, it is well-known that they share all the nonzero eigenvalues, the largest in particular, implying that once we find the eigenvector v corresponding to the largest eigenvalue of $P_B^T P_A^T$, we know that the eigenvector corresponding to the largest eigenvalue of $P_A^T P_B^T$ will be $w = P_A^T v$, and vice versa.

Equations $v = P_B^T P_A^T v$ and $w = P_A^T P_B^T w$ define the influence weight for institutions and journals, respectively. The influence weights per publication, v' and w' are:

$$v'_i = \frac{(P_B^T P_A^T v)_i}{b_i}$$
$$w'_j = \frac{(P_B^T P_A^T w)_j}{a_j}$$

where b_i and a_j are the number of articles published in the selected field and time period by institutions and journals, respectively.
JIRC will also use the Power method either for institutions or journals, since both will produce the same result:

$$v(k) = \frac{P_B^T P_A^T v^{(k-1)}}{|P_B^T P_A^T v^{(k-1)}|} \tag{15}$$

that converges to the eigenvector, v, associated with the largest eigenvalue, and the desired normalized influence vector, v'. The equation $w = P_A^T v$ will render the normalized eigenvector for the journal to journal transfer of reputation,

$$w'_j = \frac{(P_A^T v)_j}{a_j} \tag{16}$$

4 Methods

4.1 Data Collection and Preparation

Our analysis is based on publication data from the Web of Science (WoS) Core Collection, with sizable number of documents (article and review type) classified within the ESI Field Economics & Business in the year 2020. This field encompasses four WoS categories: Business, Economics, Finance, and Management.

The dataset includes:

- Journals (J): $M = 613$ journals focused on Economics & Business.
- Institutions (I): $N = 600$ institutions worldwide with a substantial number of publications in the ESI field Economics & Business.
- Published Articles (PA): More than $41,000$ articles published in 2020, classified within the Economics & Business ESI Field by the Clarivate motor InCites, including journal names and institutional affiliations.
- Citing Articles (CA): More than $200,000$ articles published between 2020 and 2025 classified within the Economics & Business ESI Field by the Clarivate motor InCites, citing any of the articles belonging to PA, including their full citation data and institutional affiliations.

4.2 Matrix Construction: The JIRC Framework

As explained before, the core of our method involves the construction of two citation matrices that form a closed loop for reputation recirculation.

- Matrix A (Institutions $\rightarrow$ Journals): an (N $\times$ M) matrix where each element a_{ij} is the total number of citations from all articles in CA published by institution i to all articles in PA published in journal j.
- Matrix B (Journals $\rightarrow$ Institutions): an (M $\times$ N) matrix where each element b_{ji} is the total number of citations from all articles in CA published in journal j to all articles published in PA by institution i.

Following the Pinski-Narin methodology, we normalize these matrices to be row-stochastic, creating transition matrices P_A and P_B (as defined in Eqs. 10 and 12 in the introduction). This normalization controls for the size of institutions and journals, ensuring we measure average influence rather than raw volume.

4.3 Solving the Reputation Recirculation System

The influence vectors for institutions, v, and journals, w, are defined by the coupled system:

$$v = P_B^T P_A^T v$$
$$w = P_A^T P_B^T w$$

To check the irreducibility of the two square matrices $P_B^T P_A^T$ and $P_A^T P_B^T$, we rely on a standard result in the field of matrix theory, connected to Perron-Frobenius theory, which states that a square matrix is irreducible if and only if its associated directed graph is strongly connected. We used an algorithm based on Dijkstra (1959) to settle the issue. Afterwards, we solve for the principal eigenvector v of the (N $\times$ N) matrix $P_B^T P_A^T$ using the power method (Eq. 15). The journal influence vector is then directly calculated as $w = P_A^T v$.

4.4 Researcher Screening Application

To validate the practical utility of JIRC in individual research evaluation, we assembled a list of 330 prominent researchers in Economics & Business. This list was split into two groups:

1. Control Group (C): 165 Researchers with universally acknowledged influence, including recipients of the Nobel Prize in Economics, the John Bates Clark Medal, Gossen Prize, and other prestigious field-specific awards, as well as European Research Council advanced grant holders.

2. Test Group (T):165 Researchers with a high number of highly-cited papers but without the prestigious awards defining the Control group.

For each paper published by any of the researchers from our sample, we computed a JIRC score as the product of the journals's JIRC score and the best institutional JIRC score among all the institution in the paper byline

Then, for each researcher, we calculated a personal JIRC score by averaging the JIRC scores of all their publications between 2015 and 2024 (matching the temporal window used by Clarivate in 2025 to identify highly cited papers). Researchers were then ranked and divided into three tiers based on this averaged score.

5 Results

5.1 Journal and Institution Rankings

The JIRC algorithm produced stable and convergent influence scores for all journals and institutions. The journal rankings ([Table/Appendix 1]) showed strong concordance with expert judgment and other reputation-based rankings, while also introducing nuance by factoring in the prestige of citing institutions. Similarly, the institutional rankings ([Table/Appendix 2]) reflected global academic reputations, effectively identifying centers of genuine intellectual influence.

5.2 Disentangling Researcher's Influence from Notoriety

The results of the researcher screening through individual JIRC scores were striking and demonstrate the core strength of the method.

- Tier 1 (Top 100 Researchers): This group was mainly composed of researchers from the Control group (88%). These are individuals whose work is not only widely cited but is cited by influential entities (journals and institutions).
- Tier 2 (Middle 130 Researchers): This group showed a mix of both Control and Test group members (75 vs 55), indicating a spectrum of influence among highly cited authors.
- Tier 3 (Bottom 100 Researchers): This group was overwhelmingly populated by researchers from the Test group (98%). This suggests that while these individuals have achieved high citation counts (notoriety or popularity), their work is cited by sources with lower aggregate influence scores, potentially indicating strategic or circular citation practices within specific sub-communities, or work that is popular but not foundational.

6 Discussion

Our proposed JIRC algorithm successfully extends the eigenvector centrality paradigm beyond the journal-journal citation network. By recirculating reputation through a journal-institution loop, we ground journal influence in the prestige of its citing institutions and, conversely, ground institutional prestige in the influence of its citing journals.

The most significant finding is the algorithm's ability to differentiate between influence and mere popularity at the individual researcher level. The clear separation between the award-winning Control group and the highly-cited-but-unawarded Test group across the tiers provides strong empirical validation. It suggests that JIRC is resistant to inflation from tactics like citation stacking or publishing in journals that prioritize rapid publication and high citation volume over intellectual rigor and influence.

The algorithm's accuracy is dependent on clean, disambiguated data for institutions. Future work will explore the impact of self-citations at the institutional and journal level, and the application of JIRC to other scientific fields and to author-level networks directly.

7 Conclusion

The Journal-Institution Reputation Circulation algorithm offers a robust and theoretically grounded framework for assessing scholarly influence. It moves beyond counting citations to weighing them based on a recursive definition of reputation that flows between producers (institutions) and disseminators (journals) of knowledge. Our results demonstrate that JIRC is not just a metric for ranking journals and institutions but also a powerful tool for research evaluation, capable of screening individual researchers to identify those with genuine, influential impact as opposed to those who have gained notoriety from a purely citation-based system. In an era of increasing publication volume and potential metric manipulation, JIRC provides a nuanced and reliable measure of true scholarly prestige.

References

Bergstrom CT, West JD, Wiseman MA (2008) The eigenfactor™ metrics. Journal of Neuroscience 28(45):11,433–11,434

Bollen J, Rodriguez M, Van de Sompel H (2006) Journal status. Scientometrics 69:669–687

Bonacich P (1987) Power and centrality: A family of measures. American Journal of Sociology 92:1170–1182, DOI https://doi.org/10.1086/228631

Brin S, Page L (1998) The anatomy of a large-scale hypertextual web search engine. Computer Networks and ISDN Systems 30(1):107–117, proceedings of the Seventh International World Wide Web Conference

Dijkstra E (1959) A note on two problems in connexion with graphs. Numerische Mathematik 1:269–271

Franceschet M (2011) Pagerank: Standing on the shoulders of giants. Communications of the ACM 54(6):92–101

Garfield E (1955) Citation indexes for science: A new dimension in documentation through association of ideas. Science 122(3159):108–111

Kleinberg JM (1999) Hubs, authorities, and communities. ACM Comput Surv 31(4es):5–es

Page L (1998) Method for node ranking in a linked database

Pinski G, Narin F (1976) Citation influence for journal aggregates of scientific publications: Theory, with application to the literature of physics. Information Processing & Management 12(5):297–312

Prathap G (2019) The pinski–narin influence weight and the ramanujacharyulu power-weakness ratio indicators revisited. Scientometrics 119(2):1173–1185, DOI 10.1007/s11192-019-03046-7

Safón V, Docampo D, Cram L (2025) Screening articles by citation reputation. Quantitative Science Studies 6:405–420

Vigna S (2016) Spectral ranking. Network Science 4(4):433–445, DOI 10.1017/nws.2016.21, URL https://dx.doi.org/10.1017/nws.2016.21

Wouters PF (1999) The citation culture. Thesis

Zitt M, Small H (2008) Modifying the journal impact factor by fractional citation weighting: The audience factor. Journal of the American Society for Information Science and Technology 59(11):1856–1860